EX. NO: 01

Date :

Study of various Network commands used in Linux & Windows.

Aim :

To study and understand the application of various networking commands used in Linux and Windows operating systems.

Basic Networking commands :

1) arp -a - ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

Output :

Interface : 192.168.56.1 ---0xb

Internet Address	physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static

2. hostname - This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output

Deepika

3. IPconfig - This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system.

Output:

Windows IP Configuration

Eaternet adapter Ethernet 2:

Connection-specific DNS Suffix:
Link-local IPv6 Address . . . . : fe80::e150:57ed:91cc:3229%11
IPv4 Address . . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :

unknown adapter McAfee VPN:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix.:

Wireless LAN adapter Local Area Connection*2:

Media state . . . . . : Media disconnected.

connection-specific DNS suffix :

Ethernet adapter Ethernet :

Media State . . . . . : Media disconnected
connection-specific DNS suffix :

Wireless LAN adapter Wi-fi :

connection-specific DNS suffix . :
Link-local IPV6 Address . . : fe80::a407:978c:4504:449e%12
IPVA Address . . . . . : 172.16.45.193
Subnet Mask . . . . . : 255.255.254.0
Default Gateway . . . . . : 172.16.44.1

4. nbtstat -a : This command helps solve problems with NetBIOS name resolution. (Nbt Stands for NetBIOS over TCP/IP).

Output :

Displays protocol statistics and current TCP/IP connections using NBT (NetBIOS over TCP/IP).

NBTSTAT [[-a RemoteName] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

-a (adapter status) lists the remote machine's name table given its name

-A (Adapter status) lists the remote machine's name table given its IP address.

-c (cache) Lists NBT's cache of remote [machine] names & their IP addresses.

-n (names) Lists local NetBIOS names.

-r (resolved) Lists names resolved by broadcast and via WINS.

-R (Reload) Purges and reloads the remote cache name table.

-S (Sessions) Lists sessions table with the destination IP addresses

-sC(sessions) Lists sessions table converting destination IP addresses to computer NETBIOS names.

-RR (ReleaseRefresh) Sends Name Release Packets to WINS & then, starts Refresh.

Remote Name — Remote host machine name

IP Address — Dotted decimal representation of the IP address.

Interval — Redisplays selected statistics, pausing Interval seconds between each displays. Press ctrl+c to stop redisplaying statistics.

5. netstat - (Network Statistics) netstat displays a statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming & outgoing as well as viewing routing tables, interface statistics etc.

Output :

Active Connections

Proto	Local Address	Foreign Address	State
TCP	172.16.45.193:49151	1.213.25.241:https	ESTABLISHED
TCP	172.16.45.193:50114	relay-f71aa327:https	ESTABLISHED
⋮	⋮	⋮	⋮

6. nslookup - (name Server lookup) is a tool used to perform DNS lookups in Linux. It is used to ~~domain~~ display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain. or the NS servers of a domain. nslookup can operate in two modes : Interactive and non-Interactive.

Output :

```
nslookup WWW.google.com

Server  : unknown
Address : 172.16.44.1

Non-authoritative answer:
Name    : WWW.google.com
Addresses : 2001:4860:4826:7700::
            2001:4860:4826:7700::
            2001:4860:4827:7700::
            2001:4860:4826:7700::
            2001:4860:482c:7700::
            2001:4860:4829a:7700::
            2001:4860:4829:7700::
            2001:4860:4828:7700::
            142.251.155.119
```

142.251.154.119
142.251.152.119
142.251.156.119
142.251.153.119.

7. pathping - pathping is unique to windows, and is basically a combination of the ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output:

usage : pathping [-g host-list] [-h maximum_hops] [-i address] [-n] [-p period] [-q num_queries] [-w timeout] [-4] [-6] target_name

Options:

-g host-list	Loose source route along host-list.
-h maximum_hops	Maximum number of hops to search for target.
-i address	use the specified source address.
-n	Do not resolve addresses to hostnames.
-p period	Wait timeout milliseconds for each reply.
-w timeout	wait period milliseconds between pings.
-4	Force using IPv4
-6	Force using IPv6.

8. Ping - (Packet INternet Grouper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to other devices.

1. # Ping hostname (Ping local host)
2. # Ping IP address (Ping 4.2.2.2)
3. # ping fully qualified domain name (Ping www.facebook.com).

Output:

usage : Ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS] [-r count] [-s count] [[-j host-list] | [-k host_list]] [-w timeout] [-R] [-S srcaddr] [-c compartment] [-P] [-4] [-6] target_name

Options:

-t Ping the specified host until stopped. To see statistics and continue - type control-Break; To stop - type control-c.

-a Resolve addresses to hostnames.

-n count Number of echo requests to send.

-l size Send buffer size.

-f Set Don't Fragment flag in packet (IPv_4-only).

-i TTL Time to Live.

-v TOS Type of service (IPv_4-only. This setting has been deprecated and has no effect on the type of service field in the IP Header).

-r count	Record route for count hops (IPv4 - only)
-s count	Time stamp for count hops (IPv4 - only)
-j host-list	Loose source route along host-list (IPv4 - only).
-k host-list	strict source route along host-list (IPv4 - only).
-w timeout	Timeout in milliseconds to wait for each reply.
-R	use routing header to test reverse route also (IPv6 - only). Per RFC 5095 the use of this routing header has been deprecated. Some systems may drop echo requests if this header is used.
-S srcaddr	source address to use.
-c compartment	Routing compartment identifier
-p	Ping a Hyper-V Network virtualization provider address.
-4	Force using IPv4.
-6	Force using IPv6.

9. Route - route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Output:

```
ROUTE [-f] [-p] [-4|-6] command [destination]
      [MASK netmask] [gateway] [METERIC
      metric] [IF interface]
```

Linux Networking Commands

1. ip

Display IP address
configure Network interface
Add / Delete IP address
Manage routing tables.

i. Display IP Address :

ip :

Display all network interface and their assigned IP address.

Op :

usage : ip [OPTIONS] object { COMMAND / help }
ip [-force] - batch filename.

1. lo : < LOOPBACK, UP, LOWER_UP > mtu 65536 qdisc noqueue state unknown group default qlen 1000.
   inet 127.0.0.1
2. enp0s3 :
   inet 10.0.2.15/24.

ii. ip link show :

Displays the status of all network interfaces.

Output :

1. lo : <LOOPBACK, UP, LOWER_UP>
2. enp0s3 : <BROADCAST, MULTICAST, LOWER_UP>

1) ip route :

Displays the routing table & default gateway.

Output:

10.0.2.2 dev enpvo83 proto dhcp src 10.0.2.15 metric 100 10.0.2.0/24 dev enpvo83 proto kernel scope link src 10.0.2.15 metric 100

2) ip address add :

Assigns a new IP address to a network interfaces.

Ex: ip address add 192.168.1.254/24

Output :

1. lo : <LOOPBACK, UP, LOWER_UP>

2. enpos3 : <BROADCAST, MULTICAST, UP, LOWER_UP>
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enpos3.

3) IPConfig :

It is a staple in many sysadmin's tool belt for configuration configuring & troubleshooting Networks.

Output :

Windows IP configuration.

unknown adapter MCAfee VPN:

Media state ..... : Media disconnected.

connection-specific DNS suffix:

Wireless LAN adapter Wi-fi 3:

Media state ..... : Media disconnected.

connection-specific DNS suffix :

4b mtr :

Combines the functionality of the ping and traceroute commands.

a> mtr google.com

shows you the statistics, including each hop with time & loss %.

Output:

Host	Loss%	Snt	Last	Avg	Best
1. 172.16.8.1	0.0%	134	0.1	0.2	0.1
2. 172.97.163.199	2.6%	134	3.9	5.2	1.1
3. ???	2.2%	138	46.6	18.6	4.1

b> mtr -g google.com

shows numeric IP address.

c> mtr -b google.com

shows numeric IP address & hostnames.

d> mtr -c 10 google.com

set the number of ping that you want.

5> tcpdump :

This command is designed for capturing & displaying packets.

i> tcpdump -i echo -c 10 host 8.8.8.8

Filter our traffic comming from a specific host.

6> Ping DEEPIKA

It display best way to test connectivity between nodes.

Syntax: ping hostname.

Output:

Pinging DEEPIKA [fe80::521f:7ba0:f461:c63%2] with 32 bytes of data:

X[Reply statistics for fe80::521f::7ba0:f461:C63%2

Ping statistics for fe80::521f::]X

Reply from fe80::521f:7ba0:f461:c63%2:time<1ms

Ping statistics for fe80::521f::7ba0:f461:c63%2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% Loss)

Approximate round trip times in mili-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

Result :

The Basic Networking & Linux commands are learned & executed successfully.